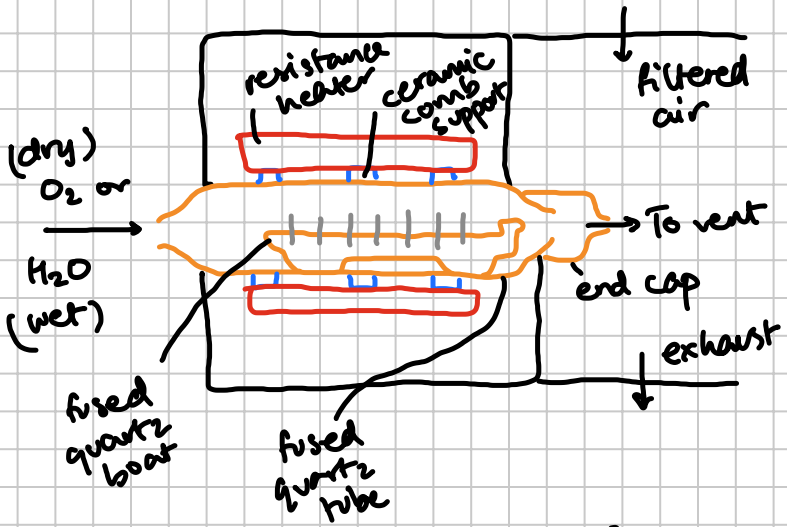
Schematic cross-section of resistance-heated oxidation furnace



$Si(s) + O_2(g) \rightarrow SiO_2(s)$ (dry)

$Si(s) + 2H_2O(g) \rightarrow SiO_2(s) + 2H_2(g)$ (wet)

T: 900°C – 1200°C

Q: 1 L/min

flow hood minimizes contamination by reducing dust + particulate matter.

controlled temp increase & decrease reqd. to prevent quartz tube + wafer warp / fracture.

∵ 1 mol of Si → 1 mol of SiO_2,

28.9 g/mol of Si → 60.08 g/mol of SiO_2

∴ $\frac{28.9}{2.33} = 12.06\ cm^3/mol$ of Si → $\frac{60.08}{2.21} = 27.18\ cm^3/mol$ of SiO_2

∴ area is constant (shared interface)

$$\frac{\text{Si thickness}}{SiO_2\ \text{thickness}} = \frac{12.06}{27.18} = 0.44$$

Thermally oxidized silicon: amorphous. (43% space occupied by SiO_2 molecules) allows impurities to diffuse.

Thermal oxidation kinetics: simple model

C_s surface concentration of oxidizing species: Co molecules/cm^3

C_o equilibrium bulk concentration @ oxidation temperature generally proportional to partial pressure of oxidant adjacent to oxide surface

oxidizing species diffuses through SiO_2 layer → C_s conc. @ Si surface

$$F_1 = D\frac{dC}{dx} \approx \frac{D(C_o - C_s)}{x}$$ D: oxidant diffusion coeff; x: thickness of oxide layer already formed.

oxidizing species chemically reacts @ silicon @ Si surface

$$F_2 = kC_s$$ assuming reaction rate ∝ C_s; k: surface reaction rate const.

At steady state, $F_1 = F_2 = F$ ∴ $F = \frac{D(C_o - C_s)}{x} = kC_s$

∴ $DC_o = DC_s + xkC_s = \frac{DkC_s}{k} + xkC_s = kC_s\left(\frac{D}{k} + x\right)$

∴ $F = \frac{DC_o}{x + (D/k)}$

of molecules of oxidizing species needed to form unit volume of oxide: C_1

$\because$ 2.2×10^{22} molecules / cm^3 in SiO_2,

C_1 — 2.2×10^{22} cm^{-3} for O_2 (1:1 reactant ratio)
↘ 4.4×10^{22} cm^{-3} for H_2O (1:2 reactant ratio)

$\therefore$ growth rate of oxide layer:

$$\frac{dx}{dt} = \frac{F}{C_1} = \frac{D C_0 / C_1}{x + (D/k)}$$

for initial condition $x(0) = d_0$ (initial oxide layer thickness)

$$(x + D/k)\,dx = D C_0 / C_1\, dt$$

$$\frac{x^2}{2} + \frac{D}{k}x = \frac{DC_0}{C_1}t + C, \quad C = \frac{d_0^2}{2} + \frac{Dd_0}{k}$$

$$\therefore \; x^2 + \frac{2Dx}{k} = \frac{2DC_0}{C_1}t + d_0^2 + \frac{2Dd_0}{k}$$

let $\tau \equiv \left(d_0^2 + \frac{2Dd_0}{k}\right)\frac{C_1}{2DC_0}$ compact: $x(t) = \frac{A}{2}\left[\sqrt{1 + \frac{4B(t+\tau)}{A^2}} - 1\right]$

$$\therefore \; x^2 + \frac{2Dx}{k} = \frac{2DC_0}{C_1}(t+\tau) \quad \text{compact: } x^2 + Ax = B(t+\tau)$$

$$\therefore \; x(t) = \frac{D}{k}\left[\sqrt{1 + \frac{2C_0 k^2 (t+\tau)}{DC_1}} - 1\right]$$

$B = \frac{2DC_0}{C_1}$

$A = \frac{2D}{k}$

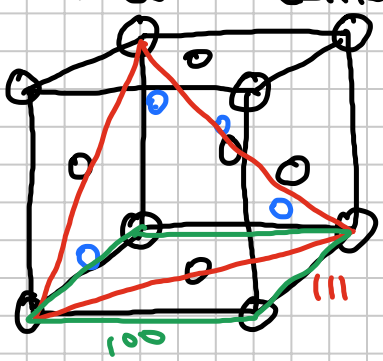
$x \cong \frac{C_0 k}{C_1}(t+\tau)$ (small t)
surface reaction limits rate
$x \propto t$
compact: $x = \frac{B}{A}(t+\tau)$ ← linear rate constant

$x \cong \sqrt{\frac{2DC_0}{C_1}(t+\tau)}$ (large t)
diffusion limits rate
$x \propto \sqrt{t}$
compact: $x^2 = B(t+\tau)$
↑ parabolic rate const

wet oxidation: d_0: very small $\tau \approx 0$ at $t = 0$

dry oxidation: $d_0 \cong 25nm$ at $t = 0$ (extrapolated value)
τ calculated from d

Diamond cubic lattice



$\ln \frac{B}{A} = -\frac{m_1}{T} + b_1$ linear const temp. dependance

$\frac{B}{A} = \exp\left(-\frac{Ea}{kT}\right)$, activation energy
$Ea \sim 2eV$ (1.83 eV to break Si-Si)
Boltzmann const.

111 Si $\frac{B}{A} > 100$ Si $\frac{B}{A}$ @ same T

bcs $\frac{B}{A}$ dependent on O_2 incorporation rate

111 bond density higher than 100.

parabolic const. temp dependance: $\ln B = -\frac{M_2}{T} + b_2$

$B = \exp\left(-\frac{E_a}{kT}\right)$ $E_a \sim$ 1.24eV (dry oxidation) 0.71eV (wet oxidation)

energy of O_2/H_2O diffusion is SiO_2 1.18eV 0.79eV

independent of crystal orientation → diffusion through amorphous layer random network.

denser & higher breakdown V

dry oxidation
↳ slower
↳ best electrical properties
↳ $\leq$ 20 nm gate oxide

wet oxidation
↳ faster
↳ worse electrical properties
↳ $\geq$ 20 nm field oxide, bipolar devices.

III oxide thickness > 100 substrate oxide thickness

for same T, t, wet oxide 5-10x thicker than dry.

Thin oxide growth — mainstream approach — 10-15nm thick layers
↳ dry O_2 ↳ atm pressure or lower ↳ lower temp 800°C - 900°C
↳ reduced O_2 partial pressure using dilwent inert gas N_2, Ar, He with gas containing oxidizing species.
↳ composite oxide films → thermally grown SiO_2 & CVD overlayer

vertical ox. furnace 10nm ± 0.1nm

rapid oxidation gives rise to $d_0 \sim 20$nm in dry ox. model above invalid.

early growth: large compressive stress in oxide reduces O_2 diffusion coeff.

$\therefore D \to 0 \Rightarrow A \to 0$. $\therefore \tau \to d_0^2/B$

$\therefore x^2 - d_0^2 = Bt$ (parabolic form w.r.t time)

Impurity redistribution during oxidation → b/w Si & SiO_2 phases.

segregation coefficient: $k = \frac{\text{Equilibrium conc. of impurity in Si}}{\text{Equilibrium conc. of impurity in } SiO_2}$

impurities may also diffuse out of SiO_2 into gaseous ambient. esp. if D of impurity in SiO_2 is large

also boundary b/w Si & SiO_2 growing into silicon: function of time relative rate of advance w.r.t. diffusion rate determines extent of redistribution.

interface trap?

masking properties for B, P, not for Ga, Al (Si_3N_4)

Oxide quality

Interface trapped charges dependent on SiO_2 - Si interface properties
energy states in Si forbidden bandgap
density dependent on crystal orientation less in 100 Si
passivated by low T (450°C) H annealing

Fixed charge within 3nm of interface. Hard to charge/discharge
generally positive, depends on oxidation & annealing cond. +
Possibly due to ionic Si or uncompleted Si orientation
charge sheets less in 100 Si Si-Si / Si-O bonds

Oxide trapped charges → SiO_2 defects removed with low T annealing

mobile ionic charges → alkali ions can cause threshold V shifts
effects reduced by adding Cl_2 (anhydrous HCl) at the cost
of higher rate constants in dry oxidation.

Thickness characterization SUPREM sim?

- comparison with reference char
- Profilometry
- Ellipsometry